

# Efficient Part-level 3D Object Generation via Dual Volume Packing

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## Abstract

Recent progress in 3D object generation has greatly improved both the quality and efficiency. However, most existing methods generate a single mesh with all parts fused together, which limits the ability to edit or manipulate individual parts. A key challenge is that different objects may have a varying number of parts. To address this, we propose a new end-to-end framework for part-level 3D object generation. Given a single input image, our method generates high-quality 3D objects with an arbitrary number of complete and semantically meaningful parts. We introduce a dual volume packing strategy that organizes all parts into two complementary volumes, allowing for the creation of complete and interleaved parts that assemble into the final object. Experiments show that our model achieves better quality, diversity, and generalization than previous image-based part-level generation methods. Our project page is at <https://research.nvidia.com/labs/dir/partpacker/>.

## 1 Introduction

Part-level 3D generation focuses on creating objects composed of multiple distinct and semantically meaningful parts, which is crucial for downstream editing and manipulation in applications such as game development and robotics. Although recent methods have greatly improved the quality of 3D object generation, they typically produce fused shapes without explicit part structures [55, 56, 59, 5, 22, 54, 57]. Generating complete and meaningful 3D parts remains a major challenge. It requires a deeper understanding of both the global layout and local part interactions within the object, which current methods still struggle to model effectively.

Several recent methods [4, 52, 51, 28] have explored part-level 3D generation by first segmenting the fused mesh into incomplete parts (usually surface patches), and then applying reconstruction or completion models to each part individually. While these pipelines have achieved promising results, they suffer from two fundamental limitations. First, they rely heavily on external segmentation priors, often derived from 2D models or pretrained networks. This introduces additional preprocessing steps and poses a risk of error propagation—any mistake in the segmentation stage can negatively impact the final generation quality. Second, these methods process each part sequentially, resulting in inefficiencies during inference. As the number of parts increases, inference time scales linearly, regardless of the individual complexity of each part. These limitations underscore two core challenges in part-level 3D generation: handling an unknown and variable number of parts, and mitigating the inefficiency of part-by-part sequential processing.

In this paper, we present a novel approach for directly generating part-level 3D objects without relying on any 2D or 3D segmentation priors. Our framework enables end-to-end generation of

\*This work is done while interning with NVIDIA.

# 通过双体积打包实现高效的部件级 3D 物体 生成

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## 摘要

近年来, 3D 物体生成领域取得了长足进展, 生成质量和效率均得到显著提升。然而, 大多数现有方法会生成一个所有部件融合在一起的单一网格, 这限制了对单个部件进行编辑或操作的能力。一个关键挑战在于, 不同物体可能包含数量各异的部件。为此, 我们提出了一种全新的端到端部件级 3D 物体生成框架。给定一张输入图像, 我们的方法能够生成具有任意数量、完整且语义明确的高质量 3D 物体部件。我们引入了一种双体积打包策略, 将所有部件组织到两个互补的体积中, 从而创建完整且交错的部件, 并将其组装成最终物体。实验表明, 与以往基于图像的部件级生成方法相比, 我们的模型在质量、多样性和泛化能力方面均表现更佳。我们的项目主页位于 <https://research.nvidia.com/labs/dir/partpacker/>。

## 1 引言

部件级三维生成专注于创建由多个不同且具有语义意义的部件组成的物体, 这对于游戏开发和机器人等应用中的后续编辑与操作至关重要。尽管近年来三维物体生成方法的质量有了显著提升, 但它们通常生成的是没有明确部件结构的融合形状 [55, 56, 59, 5, 22, 54, 57]。生成完整且具有意义的三维部件仍然是一项重大挑战。这要求深入理解物体的整体布局以及部件之间的局部交互, 而当前的方法仍难以对其进行有效建模。

近期已有一些方法 [4, 52, 51, 28] 探索部件级三维生成: 首先将融合网格分割成不完整的部件 (通常为表面片段), 然后分别对每个部件应用重建或补全模型。尽管这些流程取得了令人鼓舞的结果, 但它们存在两个根本性局限。首先, 它们高度依赖外部的分割先验, 这些先验通常来自二维模型或预训练网络。这会引入额外的预处理步骤, 并带来错误传播的风险——分割阶段的任何错误都可能对最终生成质量产生负面影响。其次, 这些方法按顺序处理每个部件, 导致推理效率低下。随着部件数量增加, 推理时间呈线性增长, 而不考虑每个部件自身的复杂程度。这些局限凸显了部件级三维生成中的两个核心挑战: 处理数量未知且可变的部件, 以及缓解逐部件顺序处理所带来的低效率问题。

本文提出了一种新颖的方法, 可在不依赖任何二维或三维分割先验的情况下, 直接生成部件级三维物体。我们的框架能够端到端地生成

\*这项工作是在英伟达实习期间完成的。



Figure 1: **End-to-end Part-level Image-to-3D Generation.** We present a method to generate 3D shape composed of individual and complete parts from a single-view image. Our method is trained only with 3D native information and can generate part-level meshes in about 30 seconds without relying on 2D segmentation prior models.

an arbitrary number of parts within a fixed time. We identify the key challenge to be the handling of overlapping regions between contacting parts. In contrast, disjoint parts, which are naturally separable as different connected components, can be processed in parallel rather than sequentially. This observation motivates our part-packing strategy, which aims to pack as many disjoint parts as possible into a shared volume to maximize space utilization, improve generation efficiency, and avoid fusing contacting parts. By analyzing the connectivity patterns commonly found in real-world objects, we observe that many can be effectively partitioned into two groups. We therefore formulate this task as a bipartite contraction problem and introduce a dual volume packing strategy, which fixes the output length and is fully compatible with existing 3D latent denoising models [56, 59].

In summary, our contributions are as follows:

1. We propose a novel end-to-end part-level 3D generation framework, which produces high-quality 3D shapes with an arbitrary number of semantically meaningful parts from a single input image, without relying on any segmentation prior.
2. We introduce a dual volume packing strategy that converts the part-connectivity graph into a bipartite graph through heuristic edge contraction, enabling efficient use of 3D volumes while preserving separation between contacting parts.
3. Experiments show that our method achieves more robust, diverse, and structurally consistent part-level generation compared to existing approaches, while offering a simpler and more efficient generation pipeline.



图 1: **端到端的部件级图像到 3D 生成。** 我们提出了一种方法, 可从单视图图像生成由独立且完整部件组成的 3D 形状。我们的方法仅使用 3D 原生信息进行训练, 无需依赖 2D 分割先验模型, 即可在约 30 秒内生成部件级网格。

固定时间内任意数量的部件。我们发现, 关键挑战在于处理相互接触部件之间的重叠区域。相比之下, 不相交部件天然可作为不同的连通分量进行分离, 因此可以并行处理, 而无需按顺序处理。这一观察促使我们提出部件打包策略, 旨在将尽可能多的不相交部件打包到共享空间中, 从而最大限度地利用空间、提高生成效率, 并避免将相互接触的部件融合。通过分析现实物体中常见的连通模式, 我们发现, 许多物体都可以有效地划分为两组。因此, 我们将该任务表述为二部图收缩问题, 并引入一种双体积打包策略。该策略固定输出长度, 并与现有的三维潜空间去噪模型完全兼容 [56, 59]。

总而言之, 我们的贡献如下:

1. 我们提出了一种新颖的端到端部件级 3D 生成框架, 可从单张输入图像生成具有任意数量语义明确部件的高质量 3D 形状, 且无需依赖任何分割先验。
2. 我们引入了一种双体积打包策略, 通过启发式边收缩将部件连接图转换为二部图, 在保持相接部件相互分离的同时, 实现对 3D 体积的高效利用。
3. 实验表明, 与现有方法相比, 我们的方法能够实现更加稳健、多样且结构一致的部件级生成, 同时提供更简单、高效的生成流程。

## 2 Related Work

### 2.1 3D Denoising Generative Models

3D-native denoising models for conditional 3D generation have seen substantial progress in recent years. Early research efforts focused on uncompressed 3D representations, such as point clouds [32], Neural Radiance Fields (NeRFs) [30, 17, 41], volumetric representations [13, 10, 33, 60, 31, 1, 2, 26, 50], and other formats [27, 3, 53, 47]. Despite their potential, these methods face limitations when applied to small or sparse datasets, often resulting in poor generalization and suboptimal quality.

More recent work has shifted towards latent denoising models for 3D generation [56, 57, 45, 21, 20, 16, 38, 9, 46, 22, 59]. These approaches typically employ a VAE to compress 3D data into a compact latent space, facilitating more efficient training of the denoising generative models. 3DShape2Vecset [55] first introduced a vector-set-based latent representation for effective 3D compression and generation. CLAY [56] demonstrates that combining VAE and 3D latent diffusion transformers yields strong performance on large-scale datasets and supports diverse conditioning modalities. TripoSG [22] pioneers the use of latent rectified flow [25, 24] and a mixture-of-experts architecture for 3D generation. Trellis [46] designs a sparse, structured latent representation with multi-format VAE decoders to support efficient rectified flow modeling. Hi3DGen [54] proposes a normal-bridge mechanism to enhance surface detail reconstruction. Dora [5] introduces salient edge sampling to improve the VAE’s reconstruction fidelity, while Hunyuan3D-2 [59, 19] explores efficient network architectures to improve generation quality and accelerate VAE decoding. In this work, we extend such 3D latent denoising models to support part-level generation.

### 2.2 Part-level 3D Generation

While existing 3D latent denoising models can generate detailed geometry, their outputs are typically fused meshes extracted from occupancy or SDF fields, where all parts are fused together. However, many practical applications require part-level 3D meshes to support editing and manipulation. A core challenge in part-level 3D generation is the varying number of parts per object, whereas latent denoising models usually rely on a fixed-length latent code, making such variability difficult to handle.

One promising direction is to employ auto-regressive models, where next-token prediction naturally accommodates variable-length outputs. For instance, MeshGPT [37] introduces this concept by tokenizing a mesh using face sorting and VQ-VAE-based compression, followed by an auto-regressive transformer to predict the token sequence. Subsequent works [6, 43, 7, 8, 14, 40, 23, 58, 42, 44] expand on this idea by designing various mesh tokenization strategies and conditioning mechanisms, including point clouds and single-view images. However, these methods often face a major limitation: complex meshes typically contain a large number of faces, making the generation process computationally expensive and time-consuming.

Another line of work focuses on segmentation-then-completion frameworks that leverage 2D segmentation prior models [18, 35]. PartGen [4] adopts a multi-view diffusion model combined with 2D segmentation to generate multi-view part segmentation maps, which are subsequently used for multi-view completion and per-part 3D reconstruction. SAMPart3D [51] proposes a zero-shot 3D part segmentation approach that transfers part-level knowledge from 2D priors to the 3D domain, enabling multi-granularity segmentation. PartField [28] learns a feed-forward 3D feature field that encodes part semantics and hierarchical structure, allowing for clustering-based segmentation and shape-level correspondence. HoloPart [52] introduces a specialized 3D latent denoising model that completes each segmented part by attending to both local geometry and global context, thereby preserving structural consistency during reconstruction. Despite their effectiveness, these methods often rely on 2D segmentation priors, which may introduce propagation errors when the input segmentations are inaccurate or inconsistent. Moreover, their pipelines tend to be complex and inefficient, typically requiring iterative processing over individual parts. To overcome these limitations, we propose an end-to-end image-to-3D generation framework that directly produces part-level 3D meshes.

## 2 相关工作

### 2.1 3D 去噪生成模型

近年来，用于条件式 3D 生成的 3D 原生去噪模型取得了长足进展。早期研究主要关注未压缩的 3D 表示形式，例如点云 [32],神经辐射场（NeRF） [30, 17, 41], 体积表示 [13, 10, 33, 60, 31, 1, 2, 26, 50], 以及其他格式 [27, 3, 53, 47]。尽管这些方法具有潜力，但应用于小规模或稀疏数据集时仍存在局限，通常会导致泛化能力较差和质量不佳。近年来，越来越多的研究转向用于 3D 生成的潜空间去噪模型 [56, 57, 45, 21, 20, 16, 38, 9, 46, 22, 59]。这些方法通常采用 VAE 将 3D 数据压缩到紧凑的潜空间中，从而实现更高效的去噪生成模型训练。3DShape2Vecset [55] 首次提出了基于向量集的潜在表示，以实现高效的 3D 压缩与生成。CLAY [56] 表明，将 VAE 与 3D 潜空间扩散 Transformer 相结合，能够在大规模数据集上取得出色性能，并支持多种条件模态。TripoSG [22]率先将潜空间整流流 [25, 24]与混合专家架构用于 3D 生成。Trellis [46] 设计了一种稀疏、结构化的潜在表示，并配备多格式 VAE 解码器，以支持高效的整流流建模。Hi3DGen [54] 提出了法线桥接机制，以增强表面细节重建效果。Dora [5] 引入显著边缘采样，以提高 VAE 的重建保真度；与此同时，Hunyuan3D-2 [59, 19] 探索高效的网络架构，以提升生成质量并加速 VAE 解码。在本文中，我们将此类 3D 潜空间去噪模型扩展到支持部件级生成。

### 2.2 部件级 3D 生成

尽管现有的 3D 潜变量去噪模型能够生成细节丰富的几何形状，但其输出通常是从占据场或 SDF 场中提取的融合网格，其中所有部件都融合在一起。然而，许多实际应用需要部件级 3D 网格，以支持编辑和操作。部件级 3D 生成面临的一个核心挑战是，不同物体所包含的部件数量各不相同，而潜变量去噪模型通常依赖固定长度的潜变量编码，因此难以处理这种变化。

一种有前景的方向是采用自回归模型，其中下一个词元的预测能够自然地适应可变量长度的输出。例如，MeshGPT [37] 通过面排序和基于 VQ-VAE 的压缩对网格进行词元化，随后使用自回归 Transformer 预测词元序列，从而引入了这一概念。后续工作 [6, 43, 7, 8, 14, 40, 23, 58, 42, 44]通过设计各种网格词元化策略和条件机制（包括点云和单视图图像）进一步拓展了这一思路。然而，这些方法通常面临一个主要局限：复杂网格往往包含大量面，使得生成过程计算成本高且耗时。

另一类研究聚焦于先分割后补全的框架，利用 2D 分割先验模型 [18, 35]。PartGen [4] 采用结合 2D 分割的多视图扩散模型，生成多视图部件分割图，随后用于多视图补全和逐部件 3D 重建。SAMPart3D [51] 提出了一种零样本 3D 部件分割方法，将 2D 先验中的部件级知识迁移到 3D 域，从而实现多粒度分割。PartField [28]学习一种前馈式 3D 特征场，对部件语义和层级结构进行编码，从而支持基于聚类的分割和形状级对应。HoloPart [52] 引入了一种专用的 3D 潜空间去噪模型，通过同时关注局部几何和全局上下文来补全每个已分割部件，从而在重建过程中保持结构一致性。尽管这些方法效果显著，但它们通常依赖 2D 分割先验；当输入分割结果不准确或不一致时，这可能引入误差传播。此外，这些方法的处理流程往往复杂且效率较低，通常需要对各个部件进行迭代处理。为克服这些局限，我们提出了一种端到端的图像到 3D 生成框架，可直接生成部件级 3D 网格。

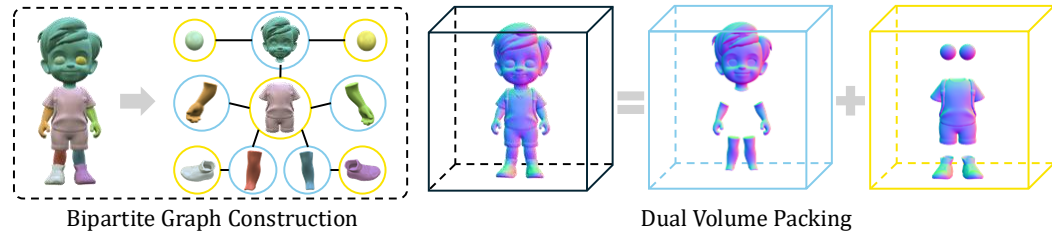


Figure 2: **Dual Volume Packing.** Given a 3D mesh with part-level annotations, we propose to convert the part-connectivity graph into a bipartite graph, such that all parts can be packed into two volumes. Within each volume, parts do not contact each other, thus can be separated during mesh extraction.

### 3 Methodology

Similar to previous 3D latent denoising models [56, 22, 59, 54, 46], our model takes a single-view image as input and generates the corresponding 3D shape. In contrast, our goal is to predict separable and complete 3D parts that can be assembled into the final shape, rather than producing a single fused mesh. We begin by introducing our dual volume packing strategy (Section 3.1) and describing the process for curating a part-level dataset (Section 3.2). We then detail the architecture of our part-level 3D generation model and the associated training procedures (Section 3.3).

#### 3.1 Dual Volume Packing

Most current 3D generation methods rely on volumetric representations, such as SDF grids, to represent shapes. Given an input image, these methods generate a single 3D volume, from which meshes can be extracted using iso-surfacing algorithms [29]. We observe that the key difficulty in part-level 3D generation lies in the handling of contacts between parts. If two parts are not in contact within the 3D space, they naturally form separate connected components in the extracted mesh and can be easily isolated. However, when parts are in contact, they become fused in the SDF grid, causing the loss of individual part information and making separation infeasible.

This observation motivates our concept of *volume packing*. **Instead of assigning each part to a separate volume, we propose packing as many disjoint parts as possible into the same volume** to maximize spatial efficiency and minimize computational cost. This problem can be formulated as a standard graph vertex coloring task. We construct a part-connectivity graph  $\mathcal{G} = \mathcal{V}, \mathcal{E}$ , where each 3D part corresponds to a vertex in  $\mathcal{V}$ , and each pair of contacting parts forms an edge in  $\mathcal{E}$ . The goal is to assign colors to vertices such that no two adjacent vertices share the same color, while minimizing the total number of colors. Parts with the same color can then be safely packed into a shared volume.

However, this strategy alone does not fully address the issue of varying output lengths. The upper bound of the chromatic number  $\chi(G)$  depends on factors such as the graph’s maximum degree and the presence of odd cycles. As illustrated in Figure 2, we further observe that many 3D objects have simple part-connectivity graphs with a bipartite structure, where  $\chi(G) = 2$ . This insight leads us to adopt a dual volume packing scheme, which is especially suitable for efficient part-level 3D latent denoising models. The output length becomes fixed, as we only need to generate two volumes, and there are only two valid colorings for each connected bipartite graph. Although it is possible to use more than two volumes, doing so introduces practical challenges: (1) When  $\chi(G) \geq 3$ , the graph often has multiple valid colorings, and the permutation invariance of color assignments complicates the learning process. (2) Using additional volumes results in lower space utilization per volume, which reduces overall efficiency. For these reasons, we adopt the dual volume packing strategy when curating our dataset.

**Bipartite Graph Extraction.** For meshes whose part-connectivity graphs are not bipartite, we apply a sequence of edge contraction operations (*i.e.*, merging specific pairs of parts) to transform the graph into a bipartite one. However, identifying the optimal set of edges to contract is known to be NP-hard [15]. We propose to use a heuristic algorithm specifically designed for the part-connectivity structures observed in meshes. Since bipartite graphs must not contain odd-length cycles, our method performs contractions to eliminate such cycles. To build the input graph, we conduct collision detection and connect an edge  $e$  between each pair of collided parts. We slightly dilate each part

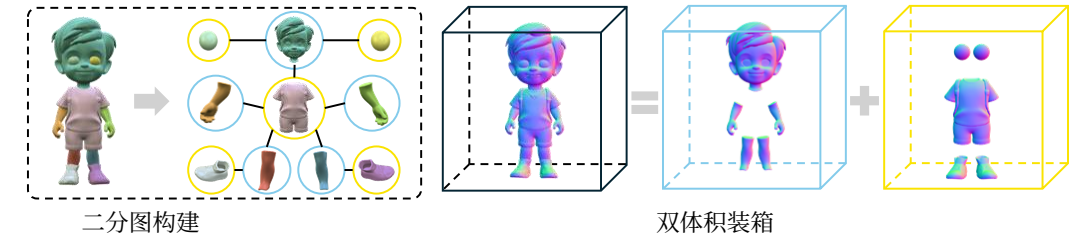


图 2: **双体积打包。** 给定一个带有部件级标注的三维网格，我们提出将部件连通图转换为二分图，使所有部件能够被打包到两个体积中。在每个体积内，部件之间彼此不接触，因此在网格提取过程中可以将其分离。

### 3 方法

与之前的 3D 潜空间去噪模型类似 [56, 22, 59, 54, 46]，我们的模型以单视图图像作为输入，并生成对应的 3D 形状。不同之处在于，我们的目标是预测可分离且完整的 3D 部件，这些部件可以组装成最终形状，而不是生成单个融合网格。我们首先介绍双体积打包策略（第 3.1 节），并描述部件级数据集的构建过程（第 3.2 节）。随后，我们详细介绍部件级 3D 生成模型的架构及其相应的训练流程（第 3.3 节）。

#### 3.1 双体积打包

当前大多数三维生成方法依赖体表示（例如 SDF 网格）来表示形状。给定一张输入图像，这些方法生成单个三维体积，然后可以使用等值面算法从中提取网格 [29]。我们观察到，部件级三维生成的关键难点在于处理部件之间的接触。如果两个部件在三维空间中不接触，它们会自然地在提取的网格中形成彼此分离的连通分量，因此很容易被单独分离出来。然而，当部件相互接触时，它们会在 SDF 网格中融合，导致单个部件的信息丢失，使分离变得不可行。

这一观察促使我们提出体积打包的概念。**我们不再将每个部件分配到单独的体积中，而是提出将尽可能多的互不相交部件打包到同一体积中**，以最大化空间利用率并降低计算成本。该问题可以表述为标准的图顶点着色任务。我们构建部件连通图  $\mathcal{G} = \mathcal{V}, \mathcal{E}$ ，其中每个三维部件对应  $\mathcal{V}$  中的一个顶点，每对相互接触的部件在  $\mathcal{E}$  中形成一条边。目标是为顶点分配颜色，使任意两个相邻顶点不共享同一种颜色，同时最小化颜色总数。颜色相同的部件随后即可安全地打包到共享体积中。

然而，仅凭这一策略仍无法完全解决输出长度不一致的问题。色数  $\chi(G)$  的上界取决于图的最大度数、奇环的存在等因素。如图 2 所示，我们进一步观察到，许多三维物体具有简单的部件连通图，并呈现二分图结构，其中  $\chi(G) = 2$ 。这一发现促使我们采用双体积打包方案，该方案尤其适用于高效的部件级三维潜空间去噪模型。由于只需生成两个体积，输出长度得以固定；对于每个连通二分图，也只有两种有效着色方式。尽管可以使用两个以上的体积，但这样做会带来实际挑战：(1) 当  $\chi(G) \geq 3$  时，图通常存在多种有效着色方式，而颜色分配的置换不变性会使学习过程更加复杂。(2) 使用额外的体积会导致每个体积的空间利用率降低，从而降低整体效率。因此，在整理数据集时，我们采用双体积打包策略。

**二分图提取。**对于部件连通图不是二分图的网格，我们会依次执行边收缩操作（即合并特定的部件对），将图转换为二分图。然而，确定最优的边收缩集合已知是 NP-hard 问题 [15]。我们提出使用一种专门针对网格中部件连通结构设计的启发式算法。由于二分图不能包含奇数长度的环，我们的方法通过收缩操作消除这类环。为构建输入图，我们进行碰撞检测，并在每对发生碰撞的部件之间连接一条边  $e$ 。我们根据 SDF 网格的分辨率对每个部件进行轻微膨胀

according to the resolution of the SDF grid to ensure that adjacent parts with tangential contact also form edges. The penetration depth between parts is used as the edge weight  $w(e)$ , with the intuition that parts with more collision should be fused. We then apply a depth-first search (DFS) algorithm to identify all cycles in the graph. To eliminate odd-length cycles, we greedily iterate over each odd cycle and contract the edge with the greatest penetration depth, effectively merging two vertices and converting the cycle into an even-length one. However, since some edges may be shared across multiple cycles, contracting a single edge can affect other cycles, potentially introducing new odd-length cycles. Therefore, we repeat this process over all cycles multiple times until no odd-length cycles remain. The full algorithm is provided in the supplementary materials.

### 3.2 Part-level Data Curation

Our packing algorithm requires 3D meshes with part-level annotations. To prepare the dataset for training, we perform part extraction, repair, and filtering to curate part-level data.

**Part Extraction.** As the 3D meshes are stored in GLB format, we extract their scene graphs and use them as the primary source of part annotations. In particular, for animated meshes, the geometry nodes typically correspond to animatable parts, serving as meaningful part annotations. However, many meshes contain only a single geometry node, with all geometric components fused into one. In such cases, we fall back on using connected components as a proxy for part annotations. Nevertheless, a semantically meaningful part may consist of multiple connected components. This issue is especially prevalent when the mesh has undergone UV unwrapping, which introduces seams in the otherwise watertight surface to flatten the 3D geometry into 2D space, resulting in multiple connected components and boundary loops. To address this, we apply a series of empirical post-processing rules: (1) We identify boundary loop pairs that share identical vertices, suggesting they originate from the same watertight surface, and merge the corresponding connected components into a single part. (2) Very small connected components are merged into an adjacent contacted part. (3) Pairs of connected components with a high intersection-over-union (IoU) score are merged into one part. While these rules do not perfectly recover ground-truth part annotations, they are sufficient for our training purposes.

**Part Repair.** When creating 3D models, artists may omit faces in occluded regions, resulting in non-watertight meshes. This issue is particularly common when meshes are separated into parts, leading to many non-watertight segments. However, VecSet-based VAEs [55] assume watertight meshes in order to learn a balanced signed distance field (SDF). A common workaround is to dilate the surface into a thin shell surrounding the geometry [56, 5], creating a watertight shape. While this technique ensures watertightness, it is undesirable because it distorts the original geometry preferred by artists and introduces unbalanced distribution of SDF values. Instead, we propose stitching certain boundary loops exposed during part extraction. This operation reduces the number of non-watertight parts and improves the balance of the resulting SDF grid for training.

**Data Filtering.** Given the two volumes of parts, ideally the amount of occupied space within each volume are roughly the same for training stability. However, some data samples may still contain extremely unbalanced distribution after the heuristic part extraction and repair processes. To ensure dataset quality, we introduce a filtering step to remove the unbalanced examples. Let  $o_1$  and  $o_2$  denote the occupancy ratios of the two packed volumes. We discard samples that satisfy the following condition:

$$((o_1 < 0.001) \wedge (o_2 < 0.001)) \vee \left( \frac{\min(o_1, o_2)}{\max(o_1, o_2)} < 0.1 \right) \quad (1)$$

This criterion effectively filters out data with highly unbalanced SDF grids, which are difficult to learn and less useful for model training.

### 3.3 Dual latent Generation

Our model builds upon several designs from previous VecSet-based latent denoising models [55], notably incorporating the salient edge sampling strategy from Dora [5] and the rectified flow model from Trellis [46]. As illustrated in Figure 3, the model is composed of three main modules. A VAE encodes the packed volumes into a compact latent code, which can later be decoded into separable mesh parts. DINOv2 [34] is used as the image encoder to extract conditional features for

以确保具有切向接触的相邻部件也能形成边。部件之间的穿透深度被用作边权重  $w(e)$ ，其直观含义是，碰撞越严重的部件越应被融合。随后，我们应用深度优先搜索（DFS）算法来识别图中的所有环。为消除奇数长度的环，我们贪心地遍历每个奇环，并收缩穿透深度最大的边，从而有效地合并两个顶点，将该环转换为偶数长度的环。然而，由于某些边可能被多个环共享，收缩一条边可能会影响其他环，并可能引入新的奇数长度环。因此，我们会对所有环重复执行该过程多次，直到不再存在奇数长度的环。完整算法见补充材料。

### 3.2 部件级数据整理

我们的包装算法需要带有部件级标注的 3D 网格。为准备训练数据集，我们通过部件提取、修复和筛选来整理部件级数据。

**部件提取。** 由于 3D 网格以 GLB 格式存储，我们提取其场景图，并将其作为部件标注的主要来源。具体而言，对于动画网格，几何节点通常对应于可动画化的部件，可作为有意义的部件标注。然而，许多网格仅包含一个几何节点，所有几何组件都融合在其中。在这种情况下，我们退而使用连通组件作为部件标注的代理。不过，一个具有语义意义的部件可能由多个连通组件组成。当网格经过 UV 展开时，这一问题尤为常见：UV 展开会在原本水密的表面上引入接缝，将 3D 几何展平到 2D 空间中，从而产生多个连通组件和边界环。为解决这一问题，我们采用了一系列经验性的后处理规则：(1) 我们识别共享完全相同顶点的边界环对，这表明它们源自同一水密表面，并将相应的连通组件合并为一个部件。(2) 将非常小的连通组件合并到相邻的接触部件中。(3) 将交并比 (IoU) 分数较高的连通组件对合并为一个部件。尽管这些规则无法完美恢复真实的部件标注，但足以满足我们的训练需求。

**部件修复。** 在创建 3D 模型时，艺术家可能会省略被遮挡区域中的面，从而导致网格不具备水密性。当网格被分割成多个部件时，这一问题尤其常见，进而产生许多非水密片段。然而，基于 VecSet 的 VAE [55] 假设网格是水密的，以便学习均衡的有符号距离场（SDF）。一种常见的解决方法是将表面膨胀为包围几何体的薄壳 [56, 5], 从而创建水密形状。虽然这种技术能够确保水密性，但它并不可取，因为会扭曲艺术家偏好的原始几何形状，并引入不平衡的 SDF 值分布。因此，我们提出在部件提取过程中，对暴露出的特定边界环进行缝合。该操作可以减少非水密部件的数量，并改善用于训练的最终 SDF 网格的均衡性。

**数据筛选。** 给定两个部件体积，理想情况下，为了保证训练稳定性，每个体积内被占据空间的大小应大致相同。然而，经过启发式部件提取和修复过程后，一些数据样本仍可能包含极不平衡的分布。为确保数据集质量，我们引入了一个筛选步骤，以移除分布不平衡的样本。令  $o_1$  和  $o_2$  表示两个打包体积的占用率。我们舍弃满足以下条件的样本：

$$((o_1 < 0.001) \wedge (o_2 < 0.001)) \vee \left( \frac{\min(o_1, o_2)}{\max(o_1, o_2)} < 0.1 \right) \quad (1)$$

该标准有效地过滤掉了 SDF 网格高度不平衡的数据，因为这类数据难以学习，对模型训练的价值也较低。

### 3.3 双潜变量生成

我们的模型建立在此前几种基于 VecSet 的潜变量去噪模型设计之上 [55]，尤其融入了 Dora 中的显著边缘采样策略 [5] 以及 Trellis 中的整流流模型 [46]。如图3所示，该模型由三个主要模块组成。VAE 将打包体编码为紧凑的潜代码，之后可将其解码为彼此分离的网格部件。DINOv2 [34] 被用作图像编码器，以提取用于

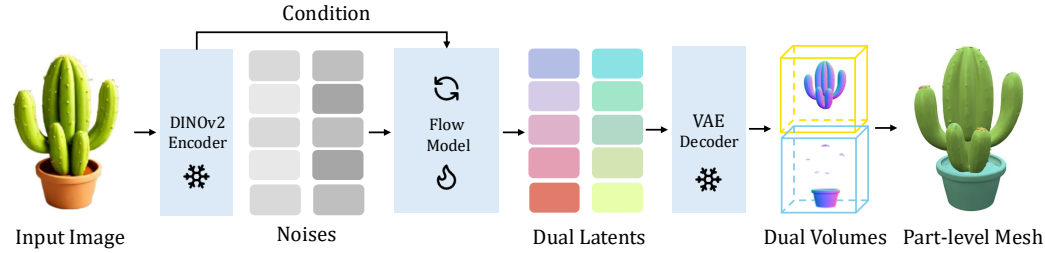


Figure 3: **Network Architecture.** Our model takes a single-view image as the input condition, and generate the dual latents at the same time with a flow model. The latents are decoded to dual volumes, which can be divided into parts and assembled back to the whole mesh.

cross-attention. Finally, a rectified flow model is trained to denoise the latent codes conditioned on these image features.

**VAE Model.** The VAE consists of a dual cross-attention encoder [5] and a self-attention decoder. Both uniform point samples and salient edge point samples are fed into the encoder. We primarily stack self-attention layers in the decoder and omit them in the encoder to accelerate the encoding process, which is invoked repeatedly during flow model training. The encoder’s intermediate features are compressed into a latent code. Following previous works [22, 59, 5], we train the VAE with multiple latent code sizes to support progressive training of the flow model and to improve convergence speed.

**Flow Model.** The flow model consists of a stack of attention layers, following similar designs as in [22, 59, 56]. A key difference is that we denoise a pair of latent codes simultaneously, rather than a single one. The two latent codes are concatenated, allowing information exchange through the attention layers. To differentiate between them, we add a learnable part embedding exclusively to the second latent code as we find this helps to reduce duplicate parts. During inference, both latent codes are predicted jointly and decoded into two separate volumes, each containing disjoint mesh parts. These parts are then assembled to reconstruct the complete 3D shape.

## 4 Experiments

### 4.1 Implementation Details

**Dataset.** We use the Trellis500k subset [46] of the Objaverse-XL dataset [12, 11] (ODC-BY v1.0 license). After applying our part extraction process (Section 3.2), approximately 386K meshes with more than one part remain. Further filtering results in around 254K meshes with well-balanced SDF grids, which are used for training our model. Specifically, we pack the extracted parts into two separate volumes and follow the data preparation pipeline established in prior work [5]: (1) Each volume is converted into a watertight mesh [56], and are then used to extract uniform surface samples, salient edge samples, and point-SDF pairs to train the VAE [5]. All meshes are normalized to the  $[-0.95, 0.95]^3$  cube, and watertight conversion is performed at a resolution of  $512^3$ , with the dilation threshold set to the voxel size. (2) Each mesh is rendered from multiple camera viewpoints to serve as the conditional input images. Following [46], we do not align the image poses with the geometry. Instead, the model is trained to learn pose-invariant generation by mapping diverse viewpoints to the canonical space.

**Training.** Our model is trained in multiple stages. We first pretrain the base VAE and flow model using the fused shape dataset without incorporating part-level information. The VAE is trained at multiple latent sizes on 64 A100 GPUs over the course of approximately one week. Next, the flow model is trained progressively with increasing latent sizes [56, 59, 5]. This progressive training phase spans about two weeks using at most 256 A100 GPUs. In the part-level stage, we observe that the VAE occasionally produces artifacts when reconstructing small parts. To mitigate this issue, we first finetune the VAE using part-level shapes. For flow model finetuning, we use the largest latent size, resulting in a total latent dimensionality of  $4096 \times 2 = 8192$ . All parameters are inherited from the pretrained model except for the newly introduced part embedding layer. The finetuning takes about two weeks using 256 A100 GPUs. Please refer to the supplementary materials for additional implementation details.

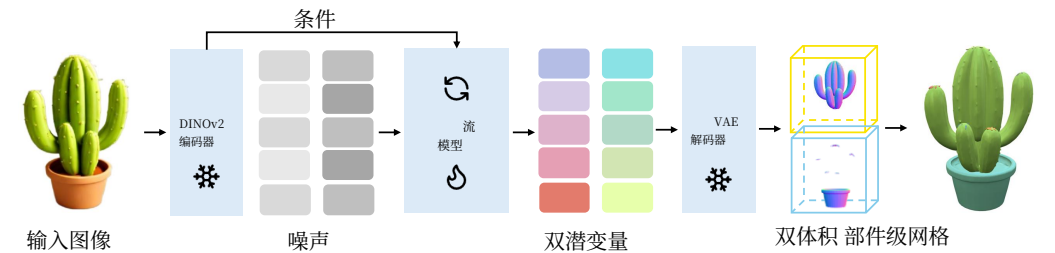


图 3: **网络架构。** 我们的模型以单视图图像作为输入条件，并通过流模型同时生成双潜变量。这些潜变量被解码为双体积，可进一步划分为各个部件，再重新组装成完整网格。

交叉注意力的条件特征。最后，训练一个整流流模型，在这些图像特征的条件下去噪。

**VAE 模型。** VAE 由一个双重交叉注意力编码器 [5] 和一个自注意力解码器组成。均匀采样的点和显著边缘上的采样点都会输入编码器。我们主要在解码器中堆叠自注意力层，而在编码器中省略这些层，以加快编码过程；在流模型训练期间，该编码过程会被反复调用。编码器的中间特征会被压缩为潜代码。继承以往工作 [22, 59, 5]，我们使用多种潜代码尺寸训练 VAE，以支持流模型的渐进式训练并提高收敛速度。

**流模型。** 流模型由多层注意力层堆叠而成，其设计与 [22, 59, 56] 中的方案类似。不同之处在于，我们会同时对一对潜代码进行去噪，而不是仅处理单个潜代码。两个潜代码会进行拼接，从而能够通过注意力层交换信息。为了区分二者，我们专门向第二个潜代码添加了可学习的部件嵌入，因为我们发现这样有助于减少重复部件。在推理过程中，两个潜代码会被联合预测，并解码为两个独立的体积，每个体积都包含互不重叠的网格部件。随后将这些部件组装起来，以重建完整的三维形状。

## 4 实验

### 4.1 实现细节

**数据集。** 我们使用 Objaverse-XL 数据集的 Trellis500k 子集 [46] [12, 11]（ODC-BY v1.0 许可证）。应用我们的部件提取流程（第 3.2 节）后，约有 386K 个包含多于一个部件的网格保留下来。进一步筛选后，得到约 254K 个 SDF 网格分布均衡的网格，用于训练我们的模型。具体而言，我们将提取出的部件打包到两个独立的体积中，并遵循先前工作所建立的数据准备流程 [5]: (1) 将每个体积转换为水密网格 [56]，然后用于提取均匀表面采样点、显著边缘采样点和点-SDF 对，以训练 VAE [5]。所有网格均归一化到  $[-0.95, 0.95]^3$  立方体，并以  $512^3$  的分辨率执行水密转换，膨胀阈值设置为体素大小。(2) 从多个相机视角渲染每个网格，作为条件输入图像。遵循 [46]，我们不对图像姿态与几何结构对齐。相反，模型通过将不同视角映射到规范空间，学习具有姿态不变性的生成。

**训练。** 我们的模型分多个阶段进行训练。首先，我们使用融合形状数据集预训练基础 VAE 和流模型，不引入部件级信息。VAE 在 64 张 A100 GPU 上以多种潜变量尺寸进行训练，持续时间约为一周。接下来，使用逐渐增大的潜变量尺寸 [56, 59, 5] 对流模型进行渐进式训练。该渐进式训练阶段历时约两周，最多使用 256 张 A100 GPU。在部件级阶段，我们观察到 VAE 在重建小部件时偶尔会产生伪影。为缓解这一问题，我们首先使用部件级形状对 VAE 进行微调。对于流模型的微调，我们采用最大的潜变量尺寸，使潜变量总维度达到  $4096 \times 2 = 8192$ 。除新引入的部件嵌入层外，所有参数均继承自预训练模型。微调使用 256 张 A100 GPU，耗时约两周。更多实现细节请参阅补充材料。

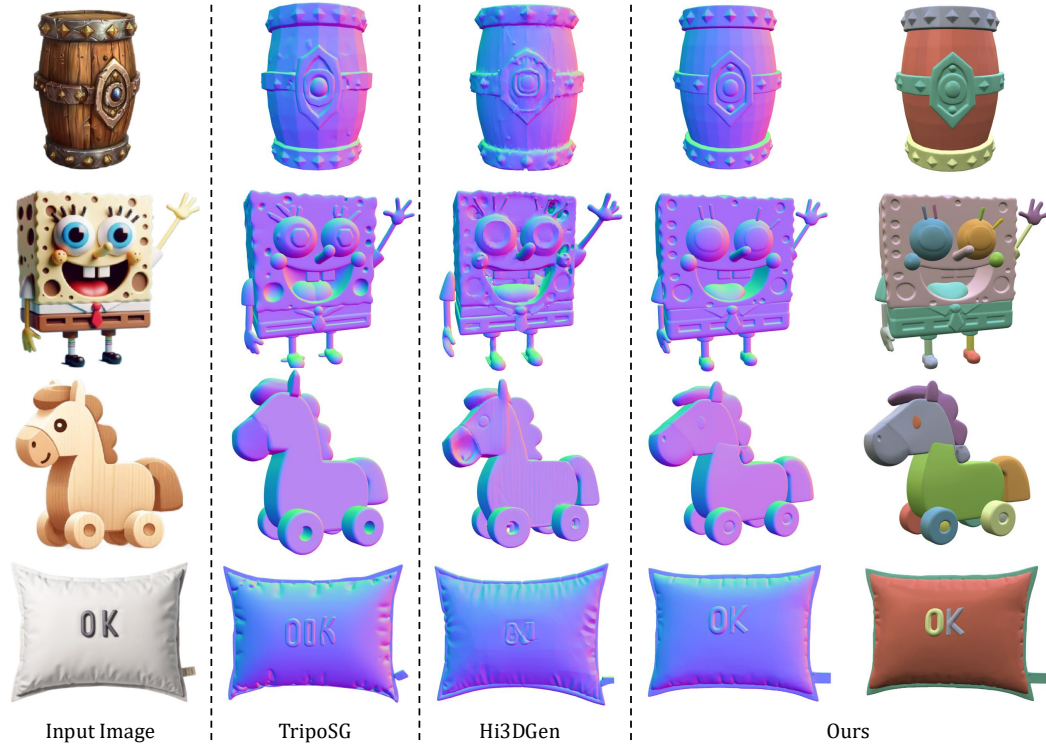


Figure 4: **Comparison on Image-to-3D Generation.** Our method generates part-level meshes with competitive quality from single-view images compared to previous methods.

#### 4.2 Qualitative Comparisons.

Our model is a latent denoising framework for image-to-3D generation, following the design principles of previous works [22, 46, 54, 59, 5]. We first evaluate the quality of 3D generation from single-view images. Figure 4 presents a qualitative comparison between our results and those generated by TripoSG [22] and Hi3DGen [54]. Our model achieves comparable or superior visual quality, demonstrating stronger alignment with the input image and producing more aesthetically pleasing mesh surfaces.

Unlike these baselines, which generate a fused shape in a single volume, our model can directly generate individual parts from a single-view image in an end-to-end manner. For part-level 3D generation, we primarily compare against HoloPart [52]. TripoSG [22] is used to generate the fused 3D mesh. We find that the original segmentation methods [51, 39] suggested by HoloPart are slow and error-prone, so we apply PartField [28] to obtain 3D segmentation masks. Following [28], we run with different cluster counts (number of parts) for each object and select the most reasonable one manually. As shown in Figure 5, our method produces more reasonable parts.

Furthermore, since our model is based on the rectified flow framework [25, 24], by varying the initial noise, we can generate diverse outputs from the same input. Figure 6 showcases examples of diverse generation results under different random seeds.

#### 4.3 Quantitative Comparisons.

We also evaluate the generation quality of image-to-3D methods from single-view images. Following Hunyuan3D-2.0 [59], we adopt ULIP [48, 49] and Uni3D [61] as evaluation metrics. These models learn unified representations across text, image, and point cloud modalities, enabling cosine similarity-based comparisons between generated 3D shapes and reference images. To perform the evaluation, we curate a test set of 40 images sourced from diverse domains, and use each method to generate corresponding 3D meshes. To ensure a fair comparison, we fix the number of denoising steps to 50, extract meshes at a grid resolution of  $512^3$ , and simplify them to 50000 faces via decimation. Since

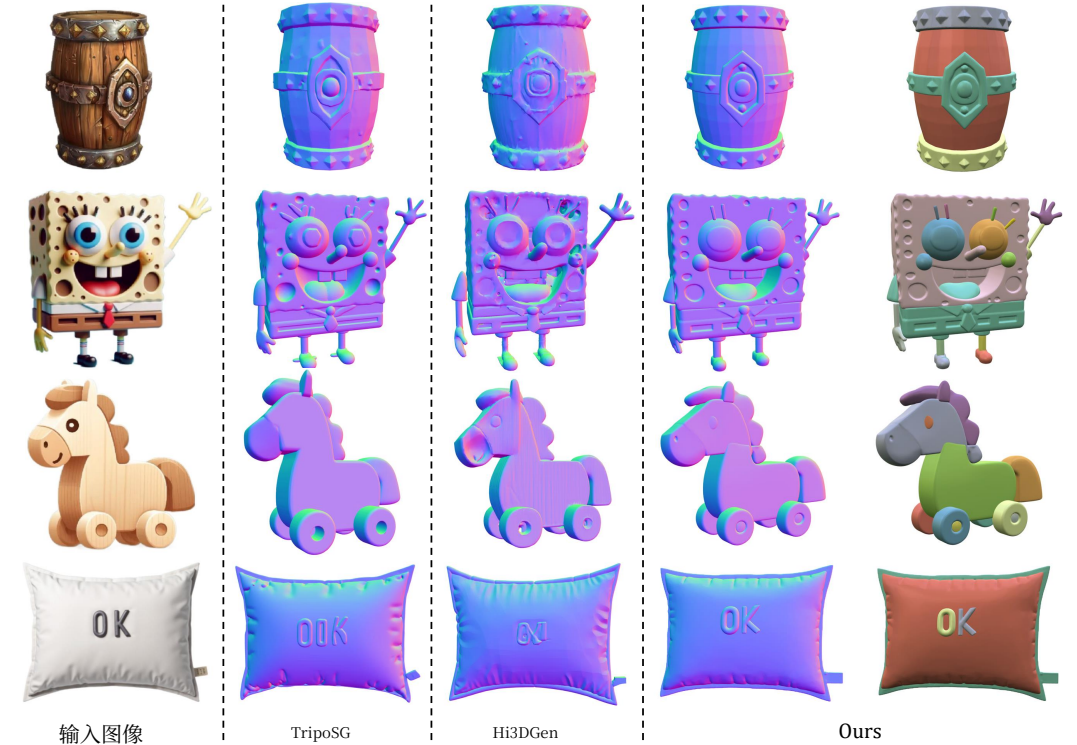


图 4: 图像到三维生成的比较。与以往方法相比, 我们的方法能够从单视图图像生成具有竞争力质量的部件级网格。

#### 4.2 定性比较。

我们的模型是一个用于图像到 3D 生成的潜变量去噪框架, 遵循先前工作的设计原则 [22, 46, 54, 59, 5]。我们首先评估单视图图像生成 3D 内容的质量。图 4 展示了我们生成的结果与 TripoSG [22]以及 Hi3DGen [54]生成结果之间的定性比较。我们的模型达到了相当甚至更优的视觉质量, 展现出与输入图像更强的对齐能力, 并生成了美观度更高的网格表面。

不同于这些基线方法将融合形状生成在单一体积中, 我们的模型能够以端到端的方式, 直接从单视图图像生成独立部件。对于部件级三维生成, 我们主要与 HoloPart [52]进行比较。TripoSG [22] 用于生成融合的三维网格。我们发现, HoloPart 提出的原始分割方法 [51, 39] 速度较慢且容易出错, 因此我们采用 PartField [28] 来获取三维分割掩码。随后, [28], 我们针对每个物体使用不同的聚类数量 (部件数量) 运行, 并手动选择最合理的结果。如图 5 所示, 我们的方法能够生成更加合理的部件。

此外, 由于我们的模型基于整流流框架 [25, 24], 通过改变初始噪声, 我们可以从同一输入生成多样化的输出。图 6 展示了不同随机种子下的多样化生成结果示例。

#### 4.3 定量比较。

我们还评估了单视图图像到 3D 方法的生成质量。继 Hunyuan3D-2 之后, 0 [59], 我们采用 ULIP [48, 49] 和 Uni3D [61] 作为评估指标。这些模型学习文本、图像和点云模态之间的统一表示, 从而能够基于余弦相似度比较生成的 3D 形状与参考图像。为进行评估, 我们整理了一个由来自不同领域的 40 张图像构成的测试集, 并使用每种方法生成相应的 3D 网格。为确保公平比较, 我们将去噪步数固定为 50, 在网格分辨率为  $512^3$  的条件下提取网格, 并通过简化将其处理为 50000 个面片。由于

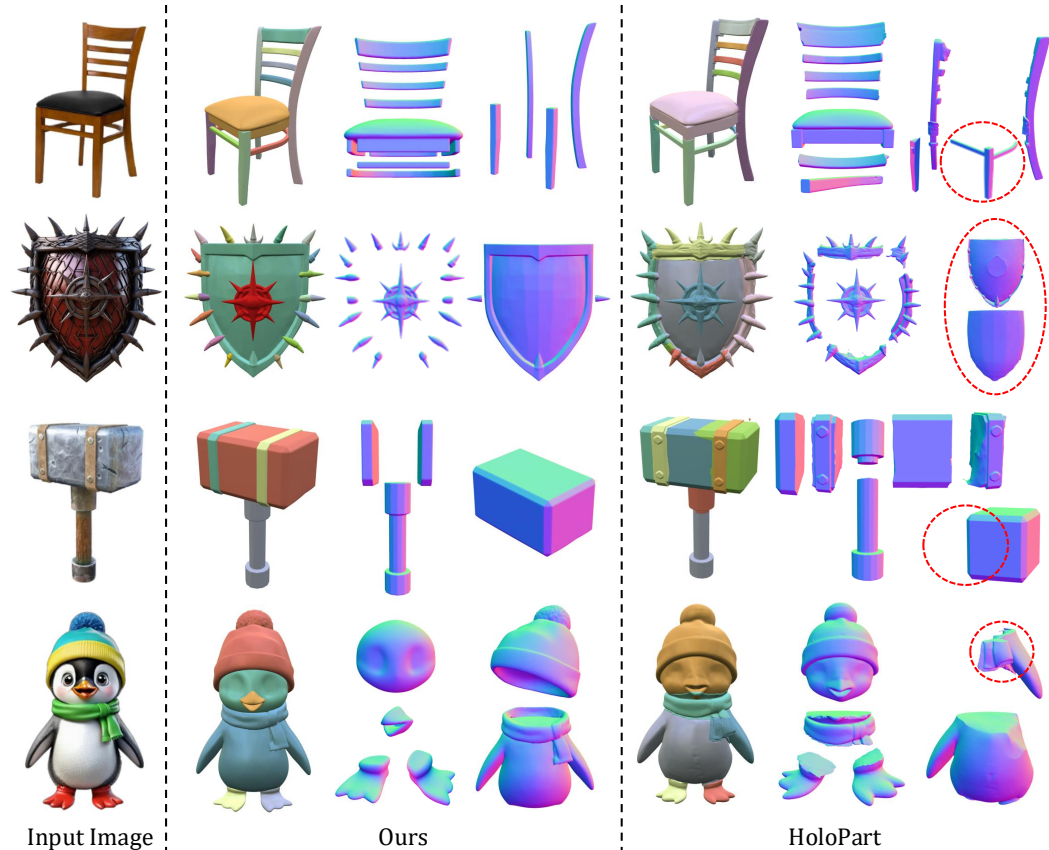


Figure 5: **Comparison on Part-level 3D Generation.** Our method directly generate complete parts, while other methods require mesh segmentation and part completion.

| Method      | Single Volume    |              |               | Dual Volumes      |        |
|-------------|------------------|--------------|---------------|-------------------|--------|
|             | Hunyuan3D-2 [59] | Hi3DGen [54] | TripoSG [22]  | Ours <sup>†</sup> | Ours   |
| ULIP [48]   | 0.1609           | 0.1641       | 0.1726        | <b>0.1729</b>     | 0.1715 |
| ULIP-2 [49] | 0.3962           | 0.3944       | <b>0.4048</b> | 0.4022            | 0.3986 |
| Uni3D [61]  | 0.3872           | 0.3864       | 0.3912        | <b>0.3941</b>     | 0.3906 |

Table 1: **Comparison of image-to-3D generation.** We measure the cosine similarity between generated meshes and input images using different feature extractors. <sup>†</sup> Our model pretrained on fused shapes to initialize our part-level model.

both ULIP-2 and Uni3D require colored point clouds as input, we assign a uniform white color to all mesh outputs before computing the metrics. As shown in Table 1, our model pretrained on fused shapes achieves competitive performance, aligning well with qualitative visual results. While the part-level fine-tuned model yields slightly lower scores due to its structural decomposition focus, it remains comparable to recent approaches, demonstrating a favorable trade-off between condition following and part-aware generation.

We further compare inference speed in the context of part-level generation. Specifically, HoloPart [52] requires a segmented mesh as input, which depends on both an image-to-3D model and a mesh segmentation model [39, 51, 28]. These two preprocessing steps alone take several minutes, and the subsequent part-wise completion time increases linearly with the number of parts. In contrast, our method takes a single-view image and directly outputs 3D parts in about 30 seconds no matter how many parts it contains, achieving huge acceleration.



图 5: **部件级 3D 生成对比。**我们的方法可直接生成完整部件，而其他方法则需要进行网格分割和部件补全。

| 方法          | 单体积              |              |               | 双体积                |        |
|-------------|------------------|--------------|---------------|--------------------|--------|
|             | Hunyuan3D-2 [59] | Hi3DGen [54] | TripoSG [22]  | 我们的方法 <sup>†</sup> | Ours   |
| ULIP [48]   | 0.1609           | 0.1641       | 0.1726        | <b>0.1729</b>      | 0.1715 |
| ULIP-2 [49] | 0.3962           | 0.3944       | <b>0.4048</b> | 0.4022             | 0.3986 |
| Uni3D [61]  | 0.3872           | 0.3864       | 0.3912        | <b>0.3941</b>      | 0.3906 |

表 1: **图像到 3D 生成的比较。**我们使用不同的特征提取器，测量生成网格与输入图像之间的余弦相似度。 <sup>†</sup> 我们的模型在融合形状上进行预训练，以初始化我们的部件级模型。

ULIP-2 和 Uni3D 都需要带颜色的点云作为输入，因此我们在计算指标之前，为所有网格输出赋予统一的白色。如表1所示，我们在融合形状上进行预训练的模型取得了具有竞争力的性能，与定性视觉结果高度一致。尽管经过部件级微调的模型由于侧重结构分解而获得的分数略低，但其性能仍与近期方法相当，展现了条件遵循能力与面向部件生成之间的良好权衡。

我们进一步比较了零件级生成场景下的推理速度。具体而言，HoloPart [52]需要以分割网格作为输入，而这同时依赖于图像到 3D 模型和网格分割模型 [39, 51, 28]。仅这两个预处理步骤就需要数分钟，后续逐零件补全的时间还会随着零件数量的增加而线性增长。相比之下，我们的方法只需输入单视图图像，无论包含多少个零件，都能在约 30 秒内直接输出 3D 零件，实现了巨大的加速。

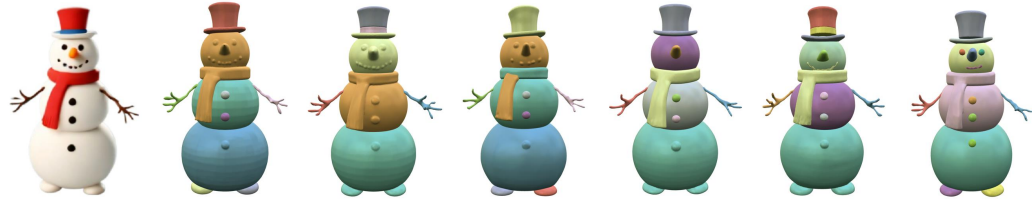


Figure 6: **Diversity on Image-to-3D Generation.** We can generate diverse and meaningful parts with different random seeds.

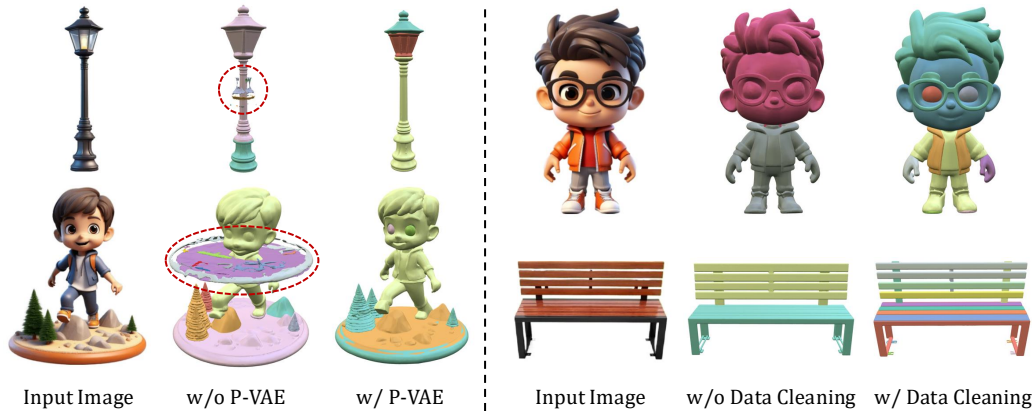


Figure 7: **Ablation Study.** We ablate different training designs and compare the generation quality.

#### 4.4 Ablation Study

We conduct ablation studies on key design choices in the part-level finetuning stage, as shown in Figure 7. First, we evaluate the impact of part-level finetuning on the VAE (denoted as P-VAE). The VAE is originally pretrained on fused shapes that are normalized and recentered; when directly applied to part-level data, which may not be centered, it often introduces artifacts. After finetuning on part-level data, the VAE becomes more robust to such spatial variations. Second, we assess the effect of the data filtering process. Without filtering, the model tends to generate overly simplistic segmentations (often consisting of only two parts), whereas filtering the training data leads to improved segmentation quality and greater diversity. Lastly, we also validate the necessity of the part embedding. We find that without adding this learnable embedding to the second latent code, the model struggles to differentiate between the dual volumes and generates overlapping or identical parts.

#### 4.5 Limitations and Future Work

Our method still exhibits several notable limitations: (1) Limited control over part granularity. The part extraction process relies solely on scene graph information from the input meshes, which are often noisy or inconsistent across the dataset. As a result, the extracted part divisions are diverse but unpredictable, hindering both consistency and user control. Incorporating additional input such as a segmentation mask may offer improved part-level controllability. (2) Constraints of bipartite contraction and dual-volume packing. While these strategies serve as practical solutions for managing complex part relationships, they are inherently limited in expressive capacity. For instance, three mutually contacting parts cannot be represented using only two volumes, leading to suboptimal results. A potential solution is to allow for more volumes, such as converting the connectivity graph into a planar graph and applying the four-color theorem [36] for packing. These limitations still constrain the model’s capacity for precise editing and joint articulation of any 3D objects.



图 6：图像到 3D 生成的多样性。我们可以通过不同的随机种子生成多样且有意义的部件。



图 7：消融研究。我们对不同的训练设计进行消融，并比较生成质量。

#### 4.4 消融研究

如图 7 所示，我们对零件级微调阶段中的关键设计选择进行了消融研究。首先，我们评估了零件级微调对 VAE（记为 P-VAE）的影响。VAE 最初是在经过归一化和重新居中的融合形状上进行预训练的；当直接应用于可能未居中的零件级数据时，往往会引入伪影。在零件级数据上进行微调后，VAE 对此类空间变化的鲁棒性更强。其次，我们评估了数据过滤过程的作用。未进行过滤时，模型倾向于生成过于简单的分割结果（通常仅包含两个零件）；而对训练数据进行过滤后，分割质量和多样性均有所提升。最后，我们还验证了零件嵌入的必要性。我们发现，如果不将这一可学习嵌入添加到第二个潜在编码中，模型便难以区分两个体积，并会生成相互重叠或完全相同的零件。

#### 4.5 局限性与未来工作

我们的方法仍存在一些值得注意的局限性：（1）对部件粒度的控制有限。部件提取过程完全依赖输入网格中的场景图信息，而这些信息在数据集内通常存在噪声或不一致性。因此，提取出的部件划分虽然多样，但难以预测，从而阻碍了一致性和用户控制。引入额外的输入信息（例如分割掩码）或许能够提升对部件级别的控制能力。（2）二部图收缩和双体积打包的约束。尽管这些策略能够有效处理复杂的部件关系，但其表达能力本质上仍然有限。例如，三个彼此接触的部件无法仅使用两个体积进行表示，从而导致次优结果。一种潜在的解决方案是允许使用更多体积，例如将连通图转换为平面图，并应用四色定理 [36] 进行打包。这些局限性仍然限制了模型对任意三维物体进行精确编辑和联合铰接的能力。

## 5 Conclusion

In this paper, we presented a novel end-to-end framework for part-level 3D generation from a single image. By leveraging 3D part connectivity and introducing a dual volume packing strategy, our method avoids reliance on 2D segmentation priors and efficiently handles an arbitrary number of parts. Experimental results demonstrate that our method achieves high-quality, diverse, and efficient part-level mesh generation, outperforming existing baselines. This work offers a promising step toward editable and structured 3D content creation for downstream applications.

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## 5 结论

本文提出了一种新颖的端到端框架，可根据单张图像生成部件级三维模型。通过利用三维部件连通性并引入双体积打包策略，我们的方法无需依赖二维分割先验，同时能够高效处理任意数量的部件。实验结果表明，我们的方法能够生成高质量、多样且高效的部件级网格，其性能优于现有基线方法。这项工作作为面向下游应用的可编辑、结构化三维内容创建迈出了有前景的一步。

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- 应当适当引用证明所依据的定理和引理。

4. 实验结果的可复现性

问题: 论文是否充分披露了复现论文主要实验结果所需的全部信息, 且披露程度足以影响论文的主要论点和/或结论 (无论是否提供代码和数据)?

回答: [是]

依据: 第 4.1 节详细说明了训练阶段、数据集筛选、潜变量大小、所使用的硬件以及数据归一化处理。

指南:

- 答案 NA 表示该论文不包含实验。
- 如果论文包含实验, 对此问题回答“否”将不会给审稿人留下好印象: 无论是否提供代码和数据, 使论文具备可复现性都很重要。
- 如果贡献是数据集和/或模型, 作者应描述为使其结果可复现或可验证而采取的步骤。

- 根据贡献的不同, 可以通过多种方式实现可复现性。例如, 如果贡献是一种新颖的架构, 完整描述该架构可能就足够了; 如果贡献是一个特定模型及其实证评估, 则可能有必要让他人能够使用相同的数据集复现该模型, 或提供模型访问权限。一般而言, 发布代码和数据通常是实现这一目标的一种好方法, 但也可以通过提供详细的结果复现说明、对托管模型的访问权限 (例如, 在大型语言模型的情况下)、发布模型检查点, 或其他适合所开展研究的方式来实现可复现性。

- 虽然 NeurIPS 不要求发布代码, 但会议确实要求所有投稿提供某种合理的可复现途径, 而具体方式可能取决于贡献的性质。例如

(a) 如果贡献主要是一种新算法, 论文应明确说明如何复现该算法。

(b) 如果贡献主要是一种新的模型架构, 论文应清晰、完整地描述该架构。

(c) 如果贡献是一个新模型 (例如大型语言模型), 那么应当提供访问该模型的方式以复现实验结果, 或者提供复现该模型的方法 (例如提供开源数据集, 或说明如何构建数据集)。

(d) 我们认识到, 在某些情况下, 可复现性可能较难实现; 在这种情况下, 欢迎作者说明其为确保可复现性所采取的具体方式。对于闭源模型, 可能只能有限地访问该模型

某种方式 (例如, 向注册用户开放), 但应当让其他研究人员能够通过某种途径复现或验证这些结果。

5. 开放获取数据和代码

Question: Does the paper provide open access to the data and code, with sufficient instructions to faithfully reproduce the main experimental results, as described in supplemental material?

Answer: [No]

Justification: The dataset is publicly available. The code will be released later.

Guidelines:

- The answer NA means that paper does not include experiments requiring code.
- Please see the NeurIPS code and data submission guidelines (<https://nips.cc/public/guides/CodeSubmissionPolicy>) for more details.
- While we encourage the release of code and data, we understand that this might not be possible, so “No” is an acceptable answer. Papers cannot be rejected simply for not including code, unless this is central to the contribution (e.g., for a new open-source benchmark).
- The instructions should contain the exact command and environment needed to run to reproduce the results. See the NeurIPS code and data submission guidelines (<https://nips.cc/public/guides/CodeSubmissionPolicy>) for more details.
- The authors should provide instructions on data access and preparation, including how to access the raw data, preprocessed data, intermediate data, and generated data, etc.
- The authors should provide scripts to reproduce all experimental results for the new proposed method and baselines. If only a subset of experiments are reproducible, they should state which ones are omitted from the script and why.
- At submission time, to preserve anonymity, the authors should release anonymized versions (if applicable).
- Providing as much information as possible in supplemental material (appended to the paper) is recommended, but including URLs to data and code is permitted.

6. Experimental setting/details

Question: Does the paper specify all the training and test details (e.g., data splits, hyper-parameters, how they were chosen, type of optimizer, etc.) necessary to understand the results?

Answer: [Yes]

Justification: Section 4.1 describes the training details.

Guidelines:

- The answer NA means that the paper does not include experiments.
- The experimental setting should be presented in the core of the paper to a level of detail that is necessary to appreciate the results and make sense of them.
- The full details can be provided either with the code, in appendix, or as supplemental material.

7. Experiment statistical significance

Question: Does the paper report error bars suitably and correctly defined or other appropriate information about the statistical significance of the experiments?

Answer: [No]

Justification: We do not report error bars or confidence intervals due to the high computational cost of repeated experiments.

Guidelines:

- The answer NA means that the paper does not include experiments.
- The authors should answer "Yes" if the results are accompanied by error bars, confidence intervals, or statistical significance tests, at least for the experiments that support the main claims of the paper.
- The factors of variability that the error bars are capturing should be clearly stated (for example, train/test split, initialization, random drawing of some parameter, or overall run with given experimental conditions).

问题：如补充材料中所述，论文是否开放提供数据和代码，并附有足够的说明，以便忠实复现主要实验结果？

回答： [否]

理由：该数据集已公开。代码将于稍后发布。

指南：

- 答案“NA”表示论文不包含需要代码的实验。
- 有关更多详细信息，请参阅 NeurIPS 代码和数据提交指南（<https://nips.cc/public/guides/CodeSubmissionPolicy>）。
- 虽然我们鼓励发布代码和数据，但我们理解这可能无法实现，因此“否”是可接受的答案。论文不会仅仅因为未包括代码，除非代码是该贡献的核心内容（例如，针对新的开源基准）。

• 说明中应包含运行所需的确切命令和环境，以便复现实验结果。详见 NeurIPS 代码和数据提交指南（<https://nips.cc/public/guides/CodeSubmissionPolicy>）。

- 作者应提供数据访问和准备方面的说明，包括如何访问原始数据、预处理数据、中间数据和生成数据等。
- 作者应提供脚本，以复现所提出的新方法和基线方法的所有实验结果。如果只能复现实验中的一部分，则应说明脚本中省略了哪些实验以及原因。

• 提交时，为了保持匿名性，作者应发布匿名版本（如适用）。

• 建议在补充材料（附于论文之后）中尽可能提供详细信息，但允许提供数据和代码的 URL。

6. 实验设置/详细信息

问题：论文是否详细说明了理解其结果所需的所有训练和测试细节（例如数据划分、超参数及其选择方式、优化器类型等）？

回答： [是]

理由：第 4.1 节 介绍了训练细节。

指南：

- 答案 NA 表示论文不包含实验。
- 实验设置应在论文主体中以足够详细的程度呈现，以便读者理解研究结果并判断其意义。

• 完整细节可以随代码提供，也可以放在附录或补充材料中。

7. 实验的统计显著性

问题：论文是否以适当且定义正确的方式报告误差线，或提供有关实验统计显著性的其他适当信息？

回答： [否]

理由：由于重复实验的计算成本很高，我们不报告误差条或置信区间。

指南：

- 答案 NA 表示该论文不包含实验。
- 如果结果附有误差线、置信区间或统计显著性检验，作者应回答“是”，至少对于支持论文主要结论的实验应如此。

• 应明确说明误差线所反映的变异因素（例如训练集/测试集划分、初始化、某些参数的随机抽取，或在给定实验条件下进行的完整运行）。

- The method for calculating the error bars should be explained (closed form formula, call to a library function, bootstrap, etc.)
- The assumptions made should be given (e.g., Normally distributed errors).
- It should be clear whether the error bar is the standard deviation or the standard error of the mean.
- It is OK to report 1-sigma error bars, but one should state it. The authors should preferably report a 2-sigma error bar than state that they have a 96% CI, if the hypothesis of Normality of errors is not verified.
- For asymmetric distributions, the authors should be careful not to show in tables or figures symmetric error bars that would yield results that are out of range (e.g. negative error rates).
- If error bars are reported in tables or plots, The authors should explain in the text how they were calculated and reference the corresponding figures or tables in the text.

8. **Experiments compute resources**

Question: For each experiment, does the paper provide sufficient information on the computer resources (type of compute workers, memory, time of execution) needed to reproduce the experiments?

Answer: [Yes]

Justification: Section 4.1 specifies the compute resources used.

Guidelines:

- The answer NA means that the paper does not include experiments.
- The paper should indicate the type of compute workers CPU or GPU, internal cluster, or cloud provider, including relevant memory and storage.
- The paper should provide the amount of compute required for each of the individual experimental runs as well as estimate the total compute.
- The paper should disclose whether the full research project required more compute than the experiments reported in the paper (e.g., preliminary or failed experiments that didn't make it into the paper).

9. **Code of ethics**

Question: Does the research conducted in the paper conform, in every respect, with the NeurIPS Code of Ethics <https://neurips.cc/public/EthicsGuidelines>?

Answer: [Yes]

Justification: The paper complies with the NeurIPS Code of Ethics.

Guidelines:

- The answer NA means that the authors have not reviewed the NeurIPS Code of Ethics.
- If the authors answer No, they should explain the special circumstances that require a deviation from the Code of Ethics.
- The authors should make sure to preserve anonymity (e.g., if there is a special consideration due to laws or regulations in their jurisdiction).

10. **Broader impacts**

Question: Does the paper discuss both potential positive societal impacts and negative societal impacts of the work performed?

Answer: [NA]

Justification: Our work focuses on foundational research in 3D object generation without immediate societal impacts, either positive or negative.

Guidelines:

- The answer NA means that there is no societal impact of the work performed.
- If the authors answer NA or No, they should explain why their work has no societal impact or why the paper does not address societal impact.

- 应说明误差棒的计算方法（闭式公式、调用库函数、自助法等）。

- 应给出所作的假设（例如，误差服从正态分布）。
- 应明确误差棒表示的是标准差还是均值的标准误。

- 报告 1-sigma 误差棒是可以的，但应予以说明。如果误差正态性的假设未经验证，作者最好报告 2-sigma 误差棒，而不是声称其具有 96% 的置信区间。

- 对于非对称分布，作者应注意不要在表格或图中显示会导致结果超出合理范围的对称误差棒（例如出现负的错误率）。

- 如果在表格或图中报告了误差棒，作者应在正文中说明其计算方法，并在正文中引用相应的图或表。

8. **实验计算资源**

问题：对于每项实验，论文是否提供了足够的计算机资源信息（计算工作节点的类型、内存、执行时间），以便复现实验？

答案： [是]

理由：第 4.1 节说明了所使用的计算资源。

指南：

- 答案 NA 表示论文未包含实验。
- 论文应说明计算工作节点的类型（CPU 或 GPU）、内部集群或云服务提供商，并包括相关的内存和存储信息。
- 论文应说明每次单独实验运行所需的计算量，并估算总计算量。

- 论文应披露完整研究项目所需的计算量是否超过论文中报告的实验所需的计算量（例如，未纳入论文的初步实验或失败实验）。

9. **伦理准则**

问题：论文中开展的研究是否在各个方面都符合 NeurIPS 伦理准则

<https://neurips.cc/public/EthicsGuidelines>？

回答： [是]

理由：论文符合 NeurIPS 伦理准则。

指南：

- 答案 NA 表示作者尚未审阅 NeurIPS 道德准则。
- 如果作者回答“否”，则应解释需要偏离道德准则的特殊情况。

- 作者应确保匿名性（例如，如果由于其司法辖区的法律或法规而存在特殊考虑）。

10. **更广泛的影响**

问题：论文是否同时讨论了所开展工作可能产生的积极社会影响和消极社会影响？

回答： [不适用]

理由：我们的工作专注于三维物体生成的基础研究，不会立即产生社会影响，无论是积极的还是消极的。

指南：

- 答案“不适用”意味着所开展工作不会产生社会影响。
- 如果作者回答“不适用”或“否”，则应解释其工作为何不会产生社会影响，或解释论文为何未涉及社会影响。

- Examples of negative societal impacts include potential malicious or unintended uses (e.g., disinformation, generating fake profiles, surveillance), fairness considerations (e.g., deployment of technologies that could make decisions that unfairly impact specific groups), privacy considerations, and security considerations.
- The conference expects that many papers will be foundational research and not tied to particular applications, let alone deployments. However, if there is a direct path to any negative applications, the authors should point it out. For example, it is legitimate to point out that an improvement in the quality of generative models could be used to generate deepfakes for disinformation. On the other hand, it is not needed to point out that a generic algorithm for optimizing neural networks could enable people to train models that generate Deepfakes faster.
- The authors should consider possible harms that could arise when the technology is being used as intended and functioning correctly, harms that could arise when the technology is being used as intended but gives incorrect results, and harms following from (intentional or unintentional) misuse of the technology.
- If there are negative societal impacts, the authors could also discuss possible mitigation strategies (e.g., gated release of models, providing defenses in addition to attacks, mechanisms for monitoring misuse, mechanisms to monitor how a system learns from feedback over time, improving the efficiency and accessibility of ML).

11. Safeguards

Question: Does the paper describe safeguards that have been put in place for responsible release of data or models that have a high risk for misuse (e.g., pretrained language models, image generators, or scraped datasets)?

Answer: [NA]

Justification: The paper does not release model or data with high misuse risk.

Guidelines:

- The answer NA means that the paper poses no such risks.
- Released models that have a high risk for misuse or dual-use should be released with necessary safeguards to allow for controlled use of the model, for example by requiring that users adhere to usage guidelines or restrictions to access the model or implementing safety filters.
- Datasets that have been scraped from the Internet could pose safety risks. The authors should describe how they avoided releasing unsafe images.
- We recognize that providing effective safeguards is challenging, and many papers do not require this, but we encourage authors to take this into account and make a best faith effort.

12. Licenses for existing assets

Question: Are the creators or original owners of assets (e.g., code, data, models), used in the paper, properly credited and are the license and terms of use explicitly mentioned and properly respected?

Answer: [Yes]

Justification: Section 4.1 states the license of the used datasets.

Guidelines:

- The answer NA means that the paper does not use existing assets.
- The authors should cite the original paper that produced the code package or dataset.
- The authors should state which version of the asset is used and, if possible, include a URL.
- The name of the license (e.g., CC-BY 4.0) should be included for each asset.
- For scraped data from a particular source (e.g., website), the copyright and terms of service of that source should be provided.
- If assets are released, the license, copyright information, and terms of use in the package should be provided. For popular datasets, [paperswithcode.com/datasets](https://paperswithcode.com/datasets) has curated licenses for some datasets. Their licensing guide can help determine the license of a dataset.

- 负面的社会影响包括潜在的恶意或非预期用途（例如，虚假信息、创建虚假个人资料、监控）、公平性考量（例如，部署可能做出不公平地影响特定群体的决策的技术）、隐私方面的考虑以及安全方面的考虑。
- 会议预计，许多论文将属于基础研究，不会与特定应用，更不用说实际部署，直接相关。然而，如果存在通向任何负面应用的直接路径，作者应当指出这一点。例如，指出生成模型质量的提升可能被用于生成用于虚假信息传播的深度伪造内容，这是合理的。另一方面，没有必要指出，通用的神经网络优化算法可能使人们能够更快地训练出生成深度伪造内容的模型。
- 作者应考虑技术在按预期使用且运行正常时可能产生的危害、技术在按预期使用但给出错误结果时可能产生的危害，以及技术被使用时可能产生的危害，但结果并不正确，以及因（有意或无意）滥用技术而产生的危害。

- 如果存在负面的社会影响，作者还可以讨论可能的缓解策略（例如，对模型进行分阶段发布；除攻击方法外同时提供防御措施；建立监测滥用行为的机制；建立监测系统如何随着时间推移从反馈中学习的机制；提高机器学习的效率和可及性）。

11. 安全措施

问题：论文是否描述了为负责任地发布具有较高滥用风险的数据或模型而采取的安全措施（例如，预训练语言模型、图像生成器或抓取的数据集）？

答案：[不适用]

理由：论文未发布具有较高滥用风险的模型或数据。

指南：

- 答案“不适用”意味着论文不存在此类风险。
- 对于滥用或双重用途风险较高的已发布模型，应在发布时采取必要的安全措施，以便对模型进行受控使用，例如要求用户遵守使用指南，限制模型访问权限，或实施安全过滤器。
- 从互联网抓取的数据集可能带来安全风险。作者应说明其如何避免发布不安全的图像。

- 我们认识到，提供有效的安全措施具有挑战性，许多论文也不要求做到这一点，但我们鼓励作者对此予以考虑，并尽最大努力采取相关措施。

12. 现有资产的许可

问题：论文中使用的资产（例如代码、数据、模型）的创作者或原始所有者是否得到了适当署名，并且是否明确注明并妥善遵守了许可协议和使用条款？

回答：[是]

理由：章节4.1说明了所用数据集的许可协议。

指南：

- 答案 NA 表示论文未使用现有资产。
- 作者应引用生成该代码包或数据集的原始论文。
- 作者应说明所使用资产的版本，并在可能的情况下附上 URL。

- 应为每项资产注明许可证名称（例如 CC-BY 4.0）。
- 对于从特定来源（例如网站）抓取的数据，应提供该来源的版权信息和服务条款。

- 如果发布资产，应提供软件包中的许可证、版权信息和使用条款。对于热门数据集，[paperswithcode.com/datasets](https://paperswithcode.com/datasets) 已为部分数据集整理了许可证信息。其许可指南有助于确定数据集的许可证。

- For existing datasets that are re-packaged, both the original license and the license of the derived asset (if it has changed) should be provided.
- If this information is not available online, the authors are encouraged to reach out to the asset’s creators.

13. **New assets**

Question: Are new assets introduced in the paper well documented and is the documentation provided alongside the assets?

Answer: [NA]

Justification: The paper does not release new assets.

Guidelines:

- The answer NA means that the paper does not release new assets.
- Researchers should communicate the details of the dataset/code/model as part of their submissions via structured templates. This includes details about training, license, limitations, etc.
- The paper should discuss whether and how consent was obtained from people whose asset is used.
- At submission time, remember to anonymize your assets (if applicable). You can either create an anonymized URL or include an anonymized zip file.

14. **Crowdsourcing and research with human subjects**

Question: For crowdsourcing experiments and research with human subjects, does the paper include the full text of instructions given to participants and screenshots, if applicable, as well as details about compensation (if any)?

Answer: [NA]

Justification: The work does not involve human subjects or crowdsourcing.

Guidelines:

- The answer NA means that the paper does not involve crowdsourcing nor research with human subjects.
- Including this information in the supplemental material is fine, but if the main contribution of the paper involves human subjects, then as much detail as possible should be included in the main paper.
- According to the NeurIPS Code of Ethics, workers involved in data collection, curation, or other labor should be paid at least the minimum wage in the country of the data collector.

15. **Institutional review board (IRB) approvals or equivalent for research with human subjects**

Question: Does the paper describe potential risks incurred by study participants, whether such risks were disclosed to the subjects, and whether Institutional Review Board (IRB) approvals (or an equivalent approval/review based on the requirements of your country or institution) were obtained?

Answer: [NA]

Justification: No human subject research is conducted.

Guidelines:

- The answer NA means that the paper does not involve crowdsourcing nor research with human subjects.
- Depending on the country in which research is conducted, IRB approval (or equivalent) may be required for any human subjects research. If you obtained IRB approval, you should clearly state this in the paper.
- We recognize that the procedures for this may vary significantly between institutions and locations, and we expect authors to adhere to the NeurIPS Code of Ethics and the guidelines for their institution.
- For initial submissions, do not include any information that would break anonymity (if applicable), such as the institution conducting the review.

- 对于重新打包的现有数据集，应提供原始许可证以及衍生资产的许可证（如果许可证已发生变化）。
- 如果无法在线获取这些信息，建议作者联系资产的创建者。

13. **新资产**

问题：论文中引入的新资产是否有完善的文档记录，并且是否随资产一并提供了文档？

回答： [不适用]

理由：该论文未发布新的资产。

指南：

- 答案 NA 表示论文未发布新资产。
- 研究人员应通过结构化模板，在投稿中说明数据集、代码或模型的详细信息，包括训练、许可、局限性等方面的信息。

- 论文应讨论是否获得了相关人员的同意，以及如何获得这些人员的同意；这些人员的资产被使用。
- 投稿时，请记得将您的资产匿名化（如适用）。您可以创建匿名 URL，或附上匿名化的 zip 文件。

14. **众包与人类受试者研究**

问题：对于涉及众包实验和人类受试者研究的论文，论文是否包含向参与者提供的完整说明文本和截图（如适用），以及有关补偿（如有）的详细信息？

回答： [不适用]

理由：该工作不涉及人类受试者或众包。

指南：

- 答案“不适用”表示该论文不涉及众包，也不涉及人类受试者研究。

- 将此信息包含在补充材料中是可以的，但如果论文的主要贡献涉及人类受试者，则应在正文中尽可能详细地说明。

- 根据 NeurIPS 道德准则，参与数据收集、整理或其他劳动的工作人员，其工资至少应达到数据收集者所在国家的最低工资标准。

15. **涉及人类受试者研究的机构审查委员会（IRB）批准或同等审批**

问题：论文是否描述了研究参与者可能面临的风险、是否向受试者披露了这些风险，以及是否获得了机构审查委员会（IRB）的批准（或根据您所在国家或机构的要求获得同等的批准/审查）？

回答： [不适用]

理由：本研究未开展涉及人类受试者的研究。

指南：

- 答案 NA 表示该论文不涉及众包，也不涉及人类受试者研究。

- 根据开展研究所在的国家，对任何人类受试者研究，可能都需要获得 IRB 批准（或同等审批）。如果您已获得 IRB 批准，应在论文中明确说明。

- 我们认识到，不同机构和地区对此的程序可能存在显著差异，并希望作者遵守 NeurIPS 道德准则以及所在机构的相关指南。

- 对于初始投稿，请勿包含任何会破坏匿名性的信息（如适用），例如开展审稿的机构。

16. **Declaration of LLM usage**

Question: Does the paper describe the usage of LLMs if it is an important, original, or non-standard component of the core methods in this research? Note that if the LLM is used only for writing, editing, or formatting purposes and does not impact the core methodology, scientific rigorousness, or originality of the research, declaration is not required.

Answer: [NA]

Justification: LLMs are not used as a core method component in this research.

Guidelines:

- The answer NA means that the core method development in this research does not involve LLMs as any important, original, or non-standard components.
- Please refer to our LLM policy (<https://neurips.cc/Conferences/2025/LLM>) for what should or should not be described.

16. **LLM 使用声明**

问题：如果大语言模型（LLM）是本研究核心方法中重要、原创或非标准的组成部分，论文是否描述了其使用情况？请注意，如果大语言模型仅用于撰写、编辑或格式设置，且未影响研究的核心方法论、科学严谨性或原创性，则无需声明。

答案： [不适用]

理由： 本研究未将 LLM 作为核心方法组成部分使用。

指南：

- 答案“不适用”意味着，本研究的核心方法开发不涉及将 LLM 作为任何重要、原创或非标准的组成部分。
- 请参阅我们的 LLM 政策（<https://neurips.cc/Conferences/2025/LLM>），了解哪些内容应该或不应该进行描述。

## A More Implementation Details

### A.1 Bipartite Contraction

We detail the greedy odd-cycle contraction algorithm used in the dual volume packing process in Algorithm 1. The algorithm first performs a depth-first search to identify all cycles (both even and odd) in the graph. It then iteratively processes each odd cycle by contracting the edge with the largest weight, repeating this step until no odd cycles remain. In Figure 8, we present the distribution of the number of parts in our processed dataset. We observe that approximately 60% of the data samples contain fewer than 10 parts. Only 5% of the samples have more than 200 parts. Since enumerating all simple cycles in a graph is not a polynomial-time operation and can be time-consuming for graphs with dense edge structures, we restrict the use of this algorithm to graphs with fewer than 100 edges to ensure processing efficiency. For more complex graphs, we directly apply two-coloring and leave the conflicting edges where both vertices have the same color as contracted edges.

---

**Algorithm 1:** Greedy Odd Cycle Contraction

---

**Data:** Graph  $\mathcal{G} = \{\mathcal{V}, \mathcal{E}\}$ , where  $\mathcal{V} = \{v_i\}_{i=1}^N$ ,  $\mathcal{E} = \{e_i\}_{i=1}^M$ .

**Result:** Array  $\mathbf{O}$  to hold the edges to contract.

```

/* Find all cycles  $\mathcal{C} = \{C_i\}$ , where each cycle  $C_i = \{e_j\}$           */
 $\mathcal{C} = \text{DFS}(\mathcal{G});$ 

/* Loop and contract until there are no odd cycles.                        */
while  $\mathcal{C}$  contains odd cycles:
    for  $C$  in  $\mathcal{C}$ :
        if  $C$  is an odd cycle:
            /* Find the edge  $e$  with the largest weight in  $C$ .          */
             $e = \arg \max_{e' \in C} w(e')$ ;
            /* Mark this edge for contraction.                          */
             $\mathbf{O}.\text{append}(e)$ ;
            /* Remove this edge from all cycles if exists.              */
            for  $C'$  in  $\mathcal{C}$ :
                 $C'.\text{remove}(e)$ ;
return  $\mathbf{O}$ ;
```

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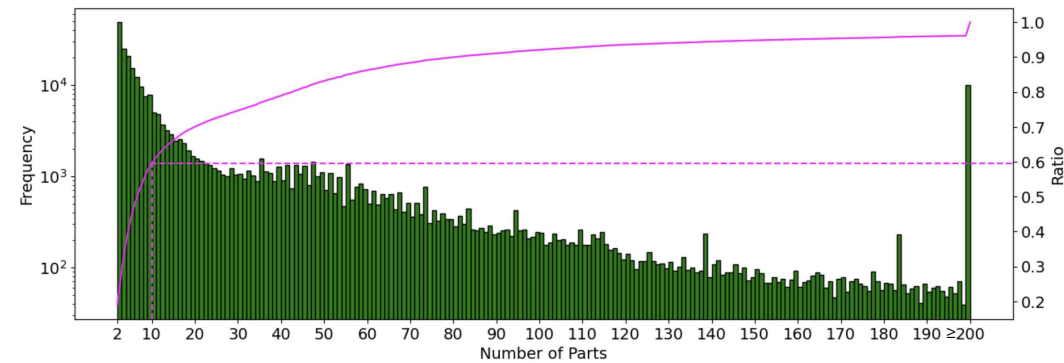


Figure 8: Statistics on the number of part in the processed dataset.

### A.2 Surface Sampling

Through dual volume packing, we split the raw mesh into two sub-meshes. For each sub-mesh, we compute the unsigned distance field (UDF) on a  $512^3$  grid. To determine the empty region (positive distance), we apply a flood-filling algorithm starting from a corner voxel (e.g., the voxel at index  $[0, 0, 0]$ ), which is guaranteed to be outside the mesh since all meshes are normalized to the range  $[-0.95, 0.95]^3$ . We then define the complement as the occupied region (negative distance), yielding a

## A 更多实现细节

### A.1 二部图收缩

我们在算法 1 中详细介绍了双重容量打包过程中所使用的贪心奇环收缩算法。该算法首先执行深度优先搜索，以识别图中的所有环（包括偶环和奇环）。随后，它迭代处理每个奇环，通过收缩权重最大的边来处理，并重复此步骤，直到不再存在奇环。在图8中，我们展示了处理后数据集中部件数量的分布情况。我们观察到，约 60% 的数据样本包含少于 10 个部件。只有 5% 的样本包含超过 200 个部件。由于枚举图中的所有简单环并非多项式时间操作，对于边结构稠密的图而言可能耗时较长，因此我们将该算法的使用范围限制在边数少于 100 条的图上，以确保处理效率。对于更复杂的图，我们直接应用二着色，并将两个顶点颜色相同的冲突边作为已收缩的边保留。

---

**Algorithm 1:** Greedy Odd Cycle Contraction

---

**Data:** Graph  $\mathcal{G} = \{\mathcal{V}, \mathcal{E}\}$ , where  $\mathcal{V} = \{v_i\}_{i=1}^N$ ,  $\mathcal{E} = \{e_i\}_{i=1}^M$ .

**Result:** Array  $\mathbf{O}$  to hold the edges to contract.

```

/* Find all cycles  $\mathcal{C} = \{C_i\}$ , where each cycle  $C_i = \{e_j\}$           */
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             $e = \arg \max_{e' \in C} w(e')$ ;
            /* Mark this edge for contraction.                          */
             $\mathbf{O}.\text{append}(e)$ ;
            /* Remove this edge from all cycles if exists.              */
            for  $C'$  in  $\mathcal{C}$ :
                 $C'.\text{remove}(e)$ ;
return  $\mathbf{O}$ ;
```

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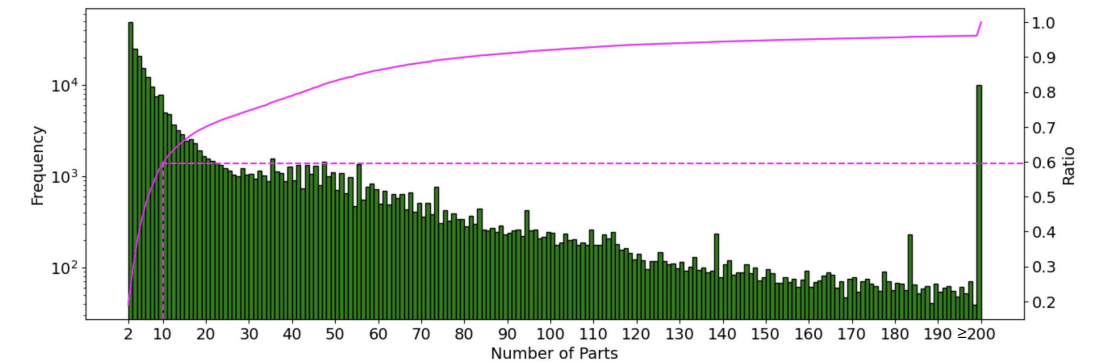


图 8：处理后数据集中部件数量的统计信息。

### A.2 表面采样

通过双体积打包，我们将原始网格拆分为两个子网格。对于每个子网格，我们在  $512^3$  网格上计算无符号距离场（UDF）。为了确定空区域（正距离），我们从一个角体素（例如索引为 $[0, 0, 0]$ 的体素）开始应用泛洪填充算法。由于所有网格都归一化到范围  $[-0.95, 0.95]^3$ ，因此可以确保该体素位于网格外部。随后，我们将其补集定义为占据区域（负距离），从而得到一个



Figure 9: More Comparisons on Image-to-3D Generation.

signed distance field. Using Marching Cubes [29], we extract a watertight surface from the resulting signed distance field.

Following Dora [5], we sample 32768 uniformly distributed surface points and 16384 salient edge points as input to the VAE. The dihedral angle threshold for salient edge detection is set to less than  $165^\circ$ . If no salient edges are present in the mesh, we randomly subsample from the uniform surface points to serve as salient edge samples. To supervise the signed distance function (SDF) output, we further generate and store three types of point-SDF pairs: (1) Uniformly sampled points in the volume  $[-1, 1]^3$ ; (2) Near-surface point samples; (3) Near-salient-edge point samples.

During VAE training, we randomly select 16384 uniform samples, 8192 near-surface samples, and 8192 near-salient-edge samples in each iteration.

## B More Results

### B.1 Qualitative Comparisons

We provide more qualitative comparisons in Figure 9. Our method outperforms prior approaches in most cases, particularly in preserving fine-grained geometric details and producing complete



图 9：图像到 3D 生成的更多比较。

有符号距离场。使用 Marching Cubes [29], 我们从得到的有符号距离场中提取出一个水密表面。

遵循 Dora [5], 我们均匀采样 32768 个表面点和 16384 个显著边缘点, 作为 VAE 的输入。显著边缘检测的二面角阈值设置为小于  $165^\circ$ 。如果网格中不存在显著边缘, 我们会从均匀表面点中随机进行子采样, 将其作为显著边缘样本。为了监督有符号距离函数 (SDF) 的输出, 我们进一步生成并存储三类点-SDF 对: (1) 体积  $[-1, 1]^3$  内均匀采样的点; (2) 近表面点样本; (3) 近显著边缘点样本。

在 VAE 训练期间, 我们在每次迭代中随机选取 16384 个均匀样本、8192 个近表面样本和 8192 个近显著边缘样本。

## B 更多结果

### B.1 定性比较

我们在图9中提供了更多定性比较结果。与以往方法相比, 我们的方法在大多数情况下表现更优, 尤其是在保留细粒度几何细节和生成完整

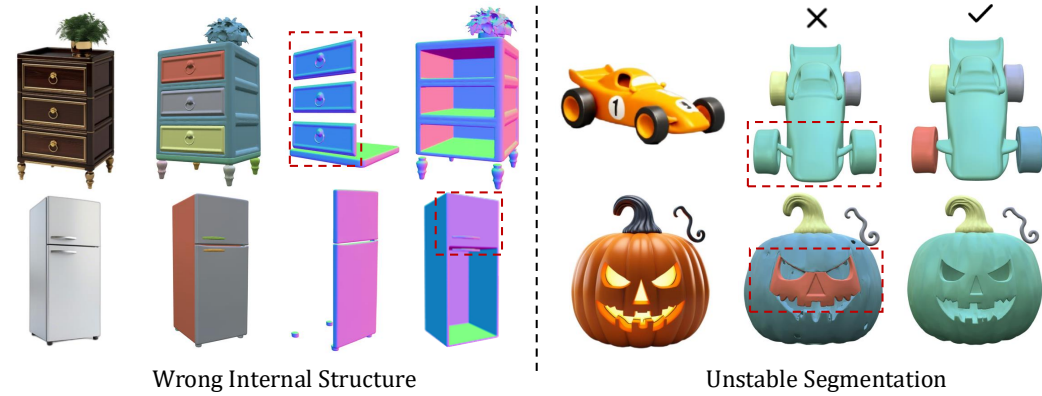


Figure 10: **Limitations and Failure Cases.** We showcase some failure cases of our model.

parts. Compared with baseline methods, our generated shapes exhibit better surface smoothness, part separation, and overall fidelity.

## B.2 Failure Cases

In Figure 10, we show some typical failure cases of our model. The first type involves incorrect internal structures. For shapes that contain intricate internal components, the underlying part connectivity graph becomes highly entangled. In such cases, our greedy odd cycle contraction algorithm may fail to partition the mesh into two meaningful groups, resulting in incomplete or incorrect reconstructions. For example, in some drawer-like objects, the inner compartment is not properly connected to the outer casing, leading to broken or missing internal parts. The second type of failure relates to unstable segmentation. Due to inconsistencies in part annotations within the training dataset and the absence of explicit part granularity control, our model can produce varying segmentation results when given different random seeds. While some samples yield satisfactory and semantically aligned part divisions, others may result in undesirable fusions or erroneous splits. For instance, the wheels of a car may sometimes be merged with the car body, violating the expected part separation. This instability highlights the need for more reliable part annotations and stronger priors to enforce consistent structural decomposition.



图 10: **局限性与失败案例。** 我们展示了模型的一些失败案例。

部件方面。与基线方法相比，我们生成的形状具有更好的表面平滑度、部件分离效果和整体保真度。

## B.2 失败案例

在图 10 中，我们展示了模型的一些典型失败案例。第一类涉及错误的内部结构。对于包含复杂内部组件的形状，其底层部件连通性图会变得高度纠缠。在这种情况下，我们的贪心奇环收缩算法可能无法将网格划分为两个有意义的组，从而导致重建不完整或不正确。例如，在一些抽屉状物体中，内部隔间未能与外部壳体正确连接，导致内部部件断裂或缺失。第二类失败与不稳定的分割有关。由于训练数据集中部件标注存在不一致，且缺乏明确的部件粒度控制，模型在使用不同随机种子时可能产生不同的分割结果。虽然一些样本能够得到令人满意且符合语义的部件划分，但另一些样本可能出现不理想的融合或错误的拆分。例如，汽车的车轮有时可能会与车身合并，违背预期的部件分离。这种不稳定性凸显了对更可靠的部件标注以及更强先验的需求，以确保一致的结构分解。